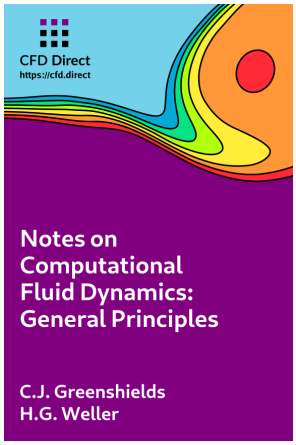
Notes on Computational Fluid Dynamics: General Principles



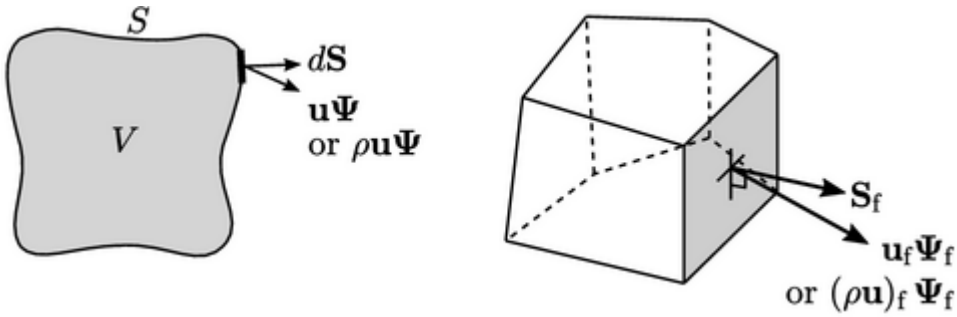
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3.9 Advection discretisation

In Sec. 2.8, we described advection terms as those of the form $\nabla \bullet (\mathbf{u}\Psi)$ or $\nabla \bullet (\rho\mathbf{u}\Psi)$. The inclusion of velocity $\mathbf{u}$ within the divergence gives the term particular characteristics that require special treatment in discretisation.



Following the finite volume principles outlined in Sec. 3.1, the discretisation approximates the surface integral by a summation over faces by

$$\nabla \bullet (\mathbf{u}\Psi) = \lim_{\Delta V \to 0} \frac{1}{\Delta V} \int_S (d\mathbf{S} \bullet \mathbf{u}\Psi) \rightarrow \frac{1}{V} \sum_f \mathbf{S}_f \bullet \mathbf{u}_f \Psi_f \rightarrow \frac{1}{V} \sum_f \phi_f \Psi_f. \tag{3.8}$$

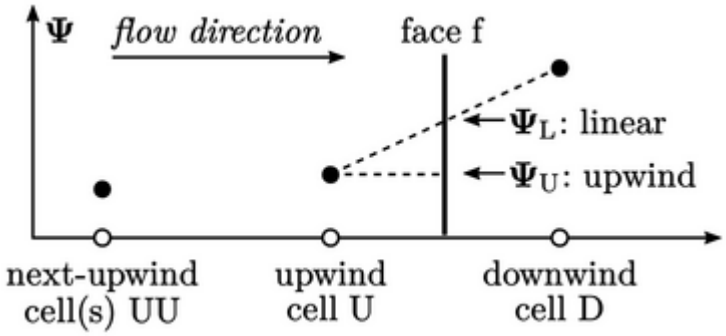
The discretisation requires calculation of the volumetric flux $\phi_f = \mathbf{S}_f \bullet \mathbf{u}_f$ (see Sec. 2.3). In the case of ρ included in the advection term $\nabla \bullet (\rho\mathbf{u}\Psi)$, ϕ_f is the mass flux $\mathbf{S}_f \bullet (\rho\mathbf{u})_f$.

In the flux calculation, the interpolation of $\mathbf{u}$ at cell centres to $\mathbf{u}_f$ at faces uses the linear scheme of Sec. 3.7. Similarly, the linear scheme is used for the interpolation $\rho\mathbf{u} \rightarrow (\rho\mathbf{u})_f$.

The critical issue — one of the most important in CFD numerics — is how to express our advected property Ψ_f at a face in terms of values Ψ in neighbouring cells.

Advection scheme introduction

The advected property Ψ is transported in the direction of flow velocity $\mathbf{u}$. Interpolation to the face of Ψ_f usually involves the flow direction. In the graphic below a face f is positioned between two cells. Based on the flow direction, the cells are labelled upwind 'U' and downwind 'D'.



The **linear** interpolation scheme, [5] described in Sec. 3.7, does not use the flow direction but expresses Ψ_f in terms of adjacent cells. At first sight, this choice of scheme is logical when considering *accuracy*. However, for advection, the linear scheme tends to generate unbounded solutions which are unstable.

The **upwind** scheme simply represents Ψ_f by the value in the upwind cell Ψ_U. It makes sense from a physical perspective since particles of fluid in the upwind cell are destined to travel to the face, transporting property Ψ_U with them.[6]

While the linear scheme is generally unbounded, the upwind scheme exhibits poor accuracy. In following section we explore the behaviour of upwind before looking at schemes that offer greater accuracy while attempting to maintain boundedness.

[5] also known as the *central difference* scheme, particularly in the context of advection.

[6] or, to quote Brian Spalding: "the wind from a pigsty always stinks".

Chris Greenshields Henry Weller 11th April 2022 Notes on CFD

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